

Mehul Gajwani, MD

Educational Background: Doctor of Medicine, UNSW 2014 – 2019

- 3rd in physiology
- 1st in Multivariable Calculus (100%)
- Cumulative biosciences mark 85%
- Practical examinations: physiology 86%; biochemistry 84%
- 5000 word thesis year 87% (high distinction)
- Preselected to represent Australia in the Inter-Medical School Physiology Quiz
- Semifinalist in the Australian Physiology Competition

Medical Officer, Northern Sydney Local District 2020 – 2021

- Registrar position in psychiatry, neurology, acute trauma/emergency (amongst other terms)

PhD: A Multiscale Approach to Mapping Brain Networks 2022 – pres

Supervised by Professor Alex Fornito and Dr James Pang, Monash University

- Research interests: network neuroscience, structure-function mapping, individual differences in brain anatomy
- Technical skills: Python (including pytorch), MATLAB, HPC (including SLURM), Unix, bash
- Awarded \$105,000 over 3 years as research allowance and \$111,000 over 3 years in tuition offset

Publications arising from PhD candidature

- Gajwani, M., Oldham, S., Pang, J., Arnatkevičiūtė, A., Tiego, J., Bellgrove, M., & Fornito, A. (2023) Can hubs of the human connectome be consistently identified with diffusion MRI? *Network Neuroscience* 7(4), 1326–1350, https://doi.org/10.1162/netn_a_00324.
- Normand, F., Gajwani, M., Côté, D. C., & Allard, A. (2023) NBS-SNI, an extension of the Network-based statistic: Abnormal functional connections between important structural actors. *Network Neuroscience* 8(1): 44–80, https://doi.org/10.1162/netn_a_00344.
- Levi, P., Chopra, S., Pang, J., Holmes, A., Gajwani, M., Sassenberg, T., DeYoung, C., & Fornito, A. (2023) The effect of using group-averaged or individualized brain parcellations when investigating connectome dysfunction in psychosis. *Network Neuroscience* 7(4): 1228–1247, https://doi.org/10.1162/netn_a_00329.
- Stein, A., Higgins, N., Gajwani, M., & Gericke, C. (2024) The Treatment Gap for Deep Brain Stimulation in Parkinson's Disease: A Comparative Analysis of Cost and Utilisation in High-Income Countries. *Australian Health Review* (in press).

Conference Talks

- Organisation of Human Brain Mapping Australia, December 2023. Invited oral presentation: *Geometric constraints on individual brain function: a deep learning approach*.
- Students of Brain Research, November 2022. Oral presentation: *The effects of different preprocessing steps and cortical parcellations on diffusion MRI connectomics*.
- Melbourne Cognitive Science Symposium, July 2022. Oral presentation: *The effects of different preprocessing steps and cortical parcellations on diffusion MRI connectomics*.
- University of Padua-Monash Connect, July 2022. Invited oral presentation: *Effects of cortical parcellation and pre-processing pipelines on diffusion MRI*.

Conference Posters

- Organization of Human Brain Mapping, July 2024. Poster presentation: *Geometric constraints on individual brain function: a deep learning approach*.
- Maths in the Brain, October 2023. Poster presentation: *Geometric constraints on individual brain function: a deep learning approach*.
- Organization of Human Brain Mapping, July 2023. Poster presentation: *Can hubs of the human connectome be consistently identified with diffusion MRI?*
- Australian Neuroscience Society, December 2022. Poster presentation: *Can hubs of the human connectome be consistently identified with diffusion MRI?*

Maths in the Brain, November 2022. Poster presentation: *Can hubs of the human connectome be consistently identified with diffusion MRI?*

Organisation of Computational Neurosciences, July 2022. Poster presentation: *The effects of different preprocessing steps and cortical parcellations on diffusion MRI connectomics.*

Community Contributions

- Honours project supervision (psychology and engineering): C. K. Boey, first class honours
- Manuscript review for *Network Neuroscience* and *Imaging Neuroscience*
- Students of Brain Research Australia: 2023 committee member
- Organisation of Human Brain Mapping and OHBM Australia: 2024 member

Prior Research Experience

Thesis Year and Visiting Scholarship

2017 – 2020

Dementia Research Unit, UNSW and Dementia Research Centre, Macquarie

- Supervised by Prof Lars Ittner and Dr Yazi Ke – Alzheimer's and FTD; publications below
- Development of novel cellular methods of healthy and pathological aging, including with lentivirus
- Development of animal models of fronto-temporal dementia and motor neuron disease
- Expertise in molecular biology, tissue culture, animal testing, microscopy and imaging

Network Visualisation Lab, Monash University

2016 – 2017

Summer Research Scholarship

- Development of software analysing high-dimensional biochemical networks
- Visualisation and analysis of system dynamics and microarray data
- Data analysis and modelling in Python, Java, and bash

Molecular Machines Group, UNSW

2015 – 2016

Summer Research Project

- MATLAB-based analysis of microscopy data; publication below

Publications

Publications prior to PhD candidature

- Stefanoska, K., Gajwani M., Tan, A., Ahel, H., Asih, P., Volkerling, A., Poljak, A., & Ittner, A. (2022) Alzheimer's disease: Ablating single master site abolishes tau hyperphosphorylation. *Science Advances* 8(27), 1-15.
- Przybyla, M., van Eersel, J., van Hummel, A., van der Hoven, J., Sabale, M., Harasta, A., Müller, J., Gajwani, M., Prikas, E., Mueller, T., Stevens, C., Power, J., Housley, G., Karl, T., Kassiou, M., Ke, Y., Ittner, A., & Ittner, L. (2020). Onset of hippocampal network aberration and memory deficits in P301S tau mice are associated with an early gene signature. *Brain (London, England : 1878)*, 143(6), 1889–1904.
- van der Hoven, J., van Hummel, A., Przybyla, M., Asih, P., Gajwani, M., Feiten, A., Ke, Y., Ittner, A., van Eersel, J., & Ittner, L. (2020). Contribution of endogenous antibodies to learning deficits and astrogliosis in human P301S mutant tau transgenic mice. *Scientific Reports*, 10(1), 13845–13845.
- Nicovich, P., Janco, M., Sobey, T., Gajwani, M., Obeidy, P., Whan, R., Gaus, K., Gunning, P., Coster, A., & Böcking, T. (2016). Effect of surface chemistry on tropomyosin binding to actin filaments on surfaces. *Cytoskeleton (Hoboken, N.J.)*, 73(12), 729–738.

References available on request

[Google Scholar](#)

[GitHub](#)

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